

The background features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

Stateful vs. stateless workloads with IaC: Together or not?

Andra Somesan

(she/her)
Solutions Architect (CEE)
AWS

Lukasz Panusz

(he/him)
Principal SA Team Lead (CEE)
AWS

Wojciech Gawronski

(he/him)
Senior Developer Advocate (CEE)
AWS

Session plan



1-hour session

10-minute intro



Open discussion

Whiteboarding focus



Survey

Materials

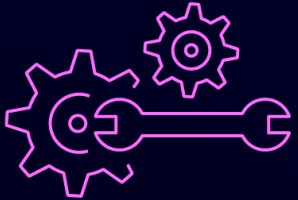


Defining **stateless** and **stateful**

What is a **state**?



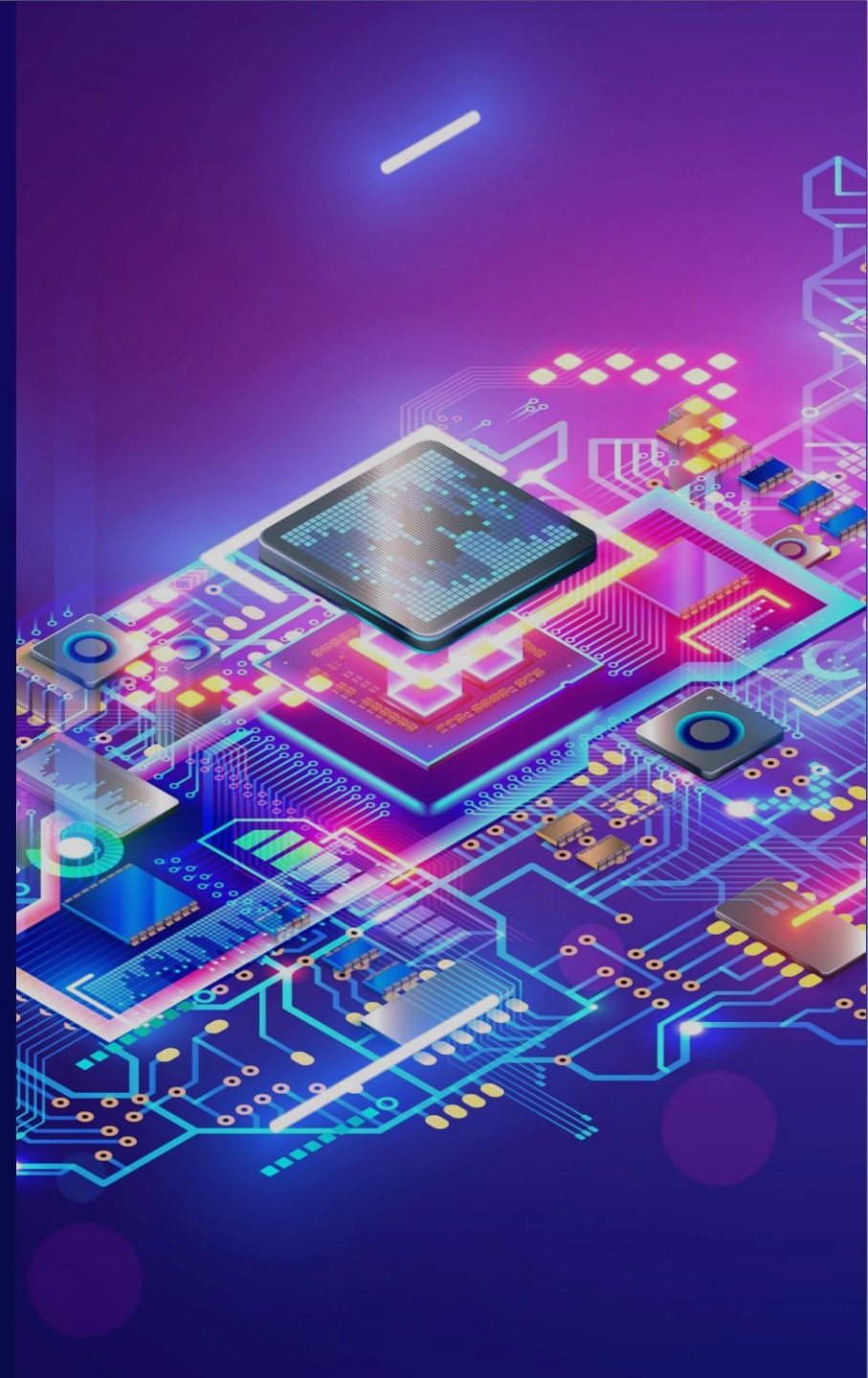
The **information** that system holds at any given point



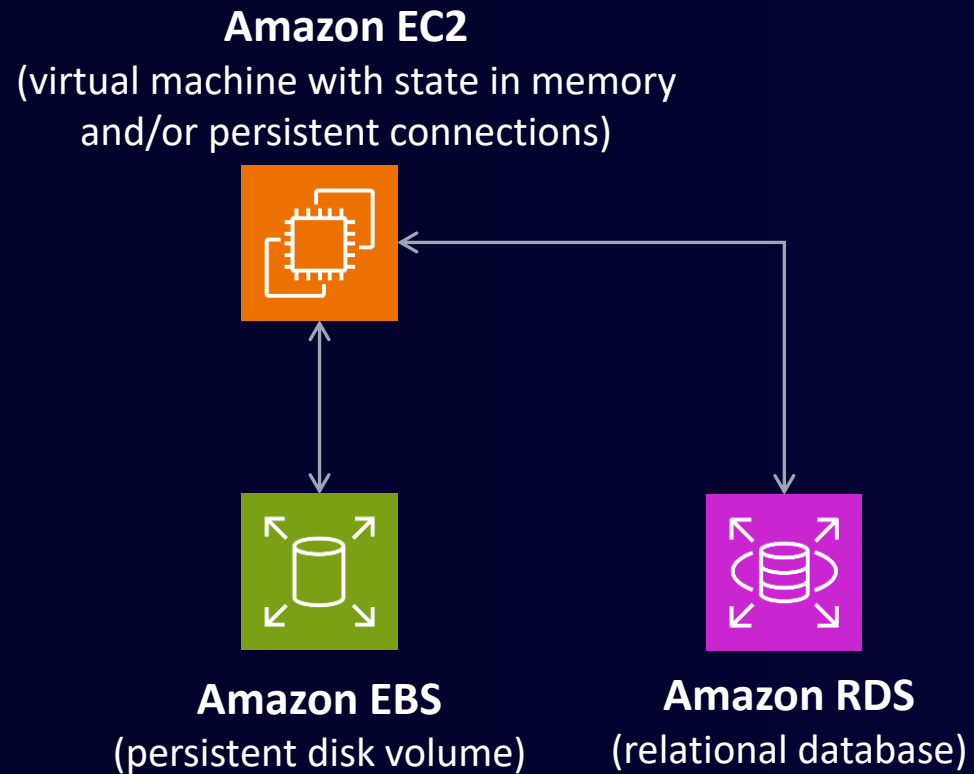
Data, configuration, user sessions, state files, etc.



Wrong resource **management** will cause side effects



What is a **side effect**?



What do we mean by **stateless**?

ALL MUTABLE OPERATIONS
(creation, modification
or destruction)

DO NOT INTRODUCE
any side effects

A replaceable resource
that ensures unaffected
operations

Interconnected infrastructure
and application components
remain undisturbed

What do we mean by **stateful**?

ALL MUTABLE OPERATIONS
(creation, modification
or destruction)

DO INTRODUCE
side effects



A **non**-replaceable resource
that remains aware of the
operations



Interconnected infrastructure
and application components
remain vulnerable
to chain effect

Stateless vs. stateful fallacies

#1: Stateful is inherently bad

#2: Transition between stateless and stateful is easy

#3: Stateful systems are always resource-heavy

#4: Stateless doesn't store data

#5: Stateless systems are inherently faster

#6: Stateful systems are outdated and not suited for modern apps

How to look at **infrastructure as code** in this context

Infrastructure as code: **Past**, **now**, and **future**

Abstractions

AWS CDK, Pulumi, Ampt, Dark

Generators

Troposphere, GoFormation

Declarative

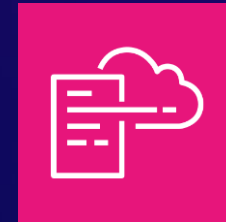
AWS CloudFormation, Terraform

Scripted/imperative

Python, Perl, Bash

Manual

Wikis, Playbooks



AWS CloudFormation



AWS Cloud
Development Kit
(AWS CDK)



HashiCorp
Terraform



Pulumi

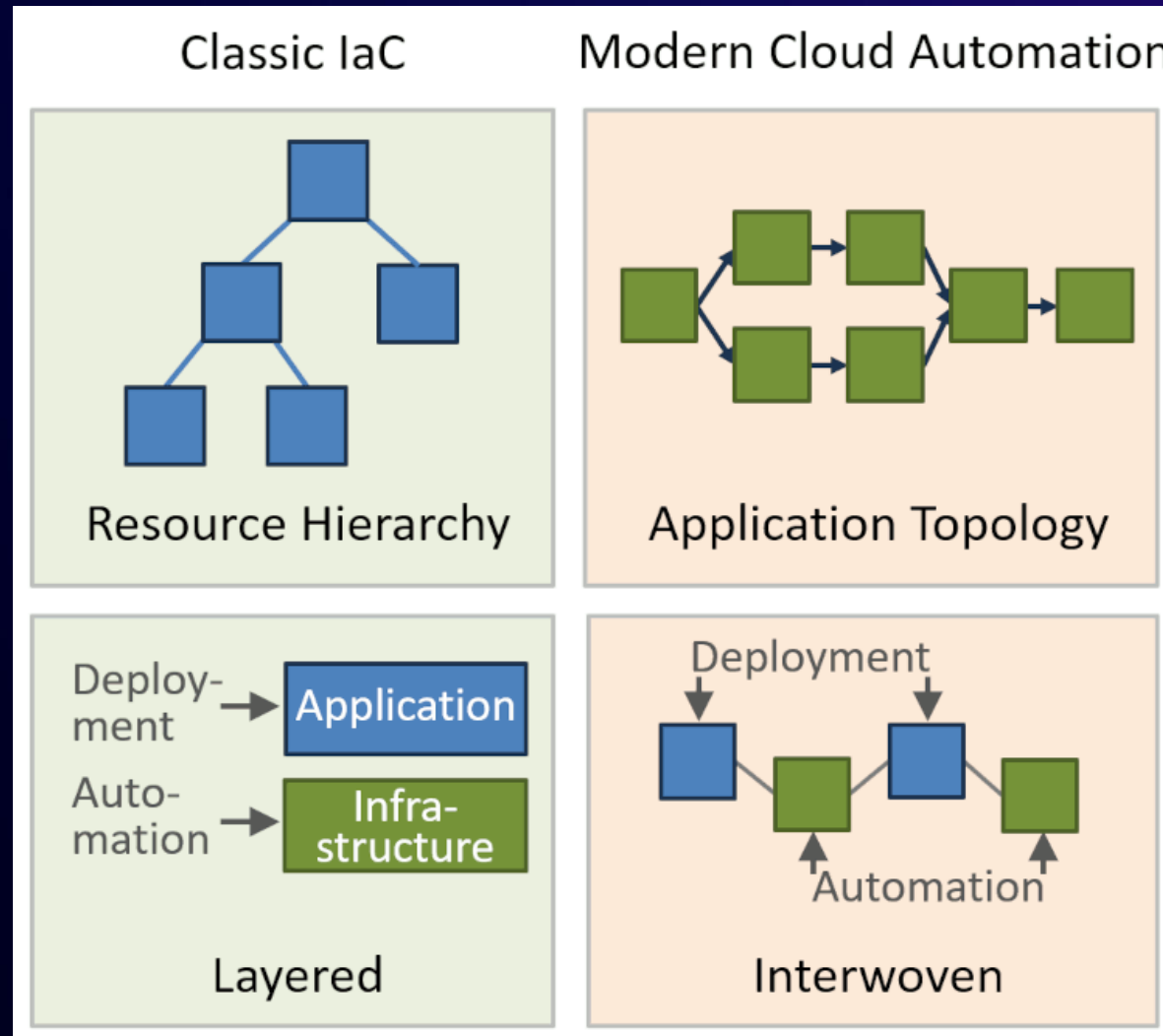




Let us clarify one thing:

Infrastructure as code is not equivalent to cloud automation, but we cannot avoid both elements in this discussion.

Relationship between IaC and cloud automation



Question:
**How do you approach stateful
workloads in IaC?**

It's time to start!



github.com/build-on-aws/stateful-workloads-in-iac

Summary



Materials



github.com/build-on-aws/stateful-workloads-in-iac

Stay connected with AWS!



Community.AWS



AWS Skill Builder



@awsdevelopers



@aws-developers



@awsdevelopers



collectives/aws

Thank you!



Please complete the session
survey in the mobile app



Andra Somesan



andrasom@amazon.com



/in/andra-somesan



Lukasz Panusz



pnszlp@amazon.com



/in/lukasz-panusz



Wojciech Gawronski



wojgaw@amazon.com



/in/afronski